REBAL: stronger baselines and protected confirmation

Experimental report | 2026-10-13 | Ten paired optimization seeds

This completed follow-up examines stronger long-tail classification controls across images, text and tabular data. Tiny ImageNet supports a fixed-treatment EQ/sampler study and an encoder/partial-tuning study; DBpedia-14 tests contextual BERT controls and selective tuning; Covertype tests weighted objectives, FT-Transformer and boosted trees. Every study retains ten paired optimization seeds, separate fit/tuning/calibration/confirmation partitions, full declared budgets, saved predictions and independent numeric verification. The primary families contain 6, 5, 8 and 3 paired macro-F1 comparisons. Per-study Holm tests are primary; a hash-bound 22-comparison global sensitivity is also retained. Results are conditional on these source partitions and model packages.

Tiny ImageNet | EQ and stronger loss controls: among the listed confirmation readouts, Selected CE + LA has the largest mean macro-F1, 56.68% (sample SD 0.10 percentage points). This ranking is descriptive; the declared paired comparisons supply the inferential evidence.

Tiny ImageNet | encoder and partial-tuning controls: among the listed confirmation readouts, Frozen DINOv2 / CE + LA has the largest mean macro-F1, 74.59% (sample SD 0.16 percentage points). This ranking is descriptive; the declared paired comparisons supply the inferential evidence.

DBpedia-14 | contextual and selective-tuning controls: among the listed confirmation readouts, Last-two-block BERT CLS / CE + LA has the largest mean macro-F1, 99.05% (sample SD 0.02 percentage points). This ranking is descriptive; the declared paired comparisons supply the inferential evidence.

Covertype | tabular model and objective controls: among the listed confirmation readouts, XGBoost + LA has the largest mean macro-F1, 93.10% (sample SD 0.05 percentage points). This ranking is descriptive; the declared paired comparisons supply the inferential evidence.

Across the 22 declared comparisons, 11 positive and 7 negative effects reach within-study exact sign-flip Holm p <= 0.05; 15 comparisons reach the separate global Holm sensitivity threshold. The complete tables retain signs, intervals and non-detections.

Every result in this report comes from a completed fixed-budget run and independent numeric replay. Fresh confirmation partitions protect this round from its own selection, but historical models may have seen these source rows. Foundation-model pretraining overlap and source dependence remain limitations.

Study	Full source	Current fit	Confirmation
Tiny ImageNet | EQ and stronger loss controls	110,000	15,268	20,000
Tiny ImageNet | encoder and partial-tuning controls	110,000	15,268	20,000
DBpedia-14 | contextual and selective-tuning controls	630,000	106,232	56,000
Covertype | tabular model and objective controls	581,012	406,704	58,110

Completed scope and evidence

Study	Tuning fits	Final fitted models	Budget
Tiny ImageNet | EQ and stronger loss controls	320	220	50 epochs/750 adapter updates; 2,000 head updates
Tiny ImageNet | encoder and partial-tuning controls	12	60	15 epochs; 1,800 AdamW updates per fit
DBpedia-14 | contextual and selective-tuning controls	26	130	50 epochs/5,200 updates or matched 3 epochs/318 updates
Covertype | tabular model and objective controls	8	40	50 epochs; MLP 9,950/FT 19,900 updates; trees 600 rounds

EQ trains 320 tuning adapters over the ten paired seeds, then reuses its selected and reference adapters for confirmation. It separately trains 220 linear classifiers for 2,000 updates each. Vision, text and tabular selection use separate dedicated tuning seeds. Counts of named readouts do not represent independent fitted models.

Each package passes its completion marker, complete file catalog hashes, frozen protocol/source identities and original independent audit. The assembler selects table rows by exact source identifiers and reruns the report builder binding checks. The saved global analysis file is hashed separately from the report labels.

The EQ audit checks all 320 adapter and 220 head states. Vision has separate 72-state and 60-model/180-readout numeric audits. Text has 156-state and 130-model/190-confirmation/100-native coverage. Tabular retains eight tuning and forty final state/budget audits and eighty numeric readouts. These are original working-run scopes; portable archives omit states and declare their narrower replay scopes.

Numeric verification recomputes metrics, global choices, decision-offset selections and paired statistics from saved predictions and labels. Actual-state audits check identities, tensors and optimizer budgets where retained. They do not independently rerun training, all encoder inference or acquisition; SHA256 binds artifacts rather than proving external replication.

Selection, confirmation and statistical inference

Configuration and checkpoint selection never optimizes confirmation macro-F1. EQ averages development scores over its ten seeds; vision uses dedicated tuning seed 901; text and tabular use seed 9. Final seed pairing and fixed source partitions are prespecified.

Calibration here selects a classification decision offset by macro-F1; the studies do not measure probability reliability. EQ and vision use calibrated tuning policies, so their calibration rows are development selection data. Text and tabular select configurations using raw tuning metrics and persist calibration decisions after global choice freeze and before confirmation inference.

The EQ amendment precedes recorded confirmation access and changes analysis roles without changing training. Primary fixed-treatment contrasts use onset .8/coefficient .05; validation-selected onset/coefficient effects remain secondary. The native source evaluation partitions remain exploratory where earlier reports informed the direction.

The sign-flip reference distribution is exhaustively enumerated over 1,024 assignments, but its inferential validity assumes paired sign exchangeability or symmetry under the null. Paired-t intervals use a normality approximation for the mean of ten optimization-seed differences, conditional on the fixed partitions. Neither rule supplies external-dataset uncertainty.

The primary endpoint is confirmation macro-F1. Accuracy, tail F1, macro recall, class-level results and distribution diagnostics remain secondary. A one percentage-point macro-F1 effect is a practical reference; it does not establish a universal deployment threshold.

Ten seeds quantify optimizer and initialization variation on fixed source partitions. Paired intervals do not include partition, entity, geographic or foundation-model pretraining uncertainty. A statistically detectable improvement can remain practically small. Non-significance does not prove equivalence.

The four prespecified primary families contain 6 EQ, 5 vision, 8 text and 3 tabular comparisons. Per-study Holm correction is primary. A separately labeled global Holm sensitivity adjusts all 22 comparisons. These corrections remain valid without assuming independence among hypotheses.

Tiny ImageNet | EQ and stronger loss controls | methodology

The full cached image source contains 100,000 development and 10,000 official-evaluation images. EQ uses the canonicalized development partitions and does not score the official 10,000-image endpoint in this round. The 110,000-source count therefore includes an unused evaluation source; canonical development is reported separately.

This GAN-free study holds the frozen DeiT feature cache fixed and compares EQ, sampling and stronger loss controls. Natural and class-uniform factorial sampling target the same expected class-balanced cross-entropy from epoch one. Natural samples receive importance weight N/(C*n_c); class-uniform samples receive weight one.

The same expected loss does not imply identical stochastic optimizer updates: sample variance, momentum and clipping can differ. The fixed primary EQ treatment begins at fraction 0.8 of training and uses coefficient 0.05 in both samplers, with matched learning rate 0.05.

CE + logit adjustment, Balanced Softmax, and a BCL adaptation use six prespecified candidates each. BCL implements class-averaged contrastive denominators and all-class classifier-weight prototypes with two adapter-dropout views of cached vectors. It is a frozen-feature adaptation, not a reproduction of BCL image augmentation or its complete original architecture.

Partition	Rows	Use
Canonical development	99973	Deduplicate and exclude official-evaluation content matches before splitting
Fit pool before thinning	71973	Separate pool; nominal 100:1 thinning follows holdout construction
Imbalanced fit	15268	Gradient updates; fit-only feature standardization
Tuning	4000	Global candidate selection across ten seeds
Calibration	4000	Decision-offset selection; also used in CE tuning policy
Confirmation	20000	100 rows per class; accessed after selection freeze

The 32 adapter configurations each run all ten seeds, giving 320 complete 50-epoch fits and 750 updates per adapter. Mean adjusted tuning macro-F1 selects CE, while the other rules use their declared effective priors. Calibration is development selection data, not untouched data reserved until after global configuration freeze.

Each EQ sampler searches six onset/coefficient settings; CE and Balanced Softmax each search six learning rates; BCL searches six coefficient/temperature settings. EQ-off factorial controls remain fixed and matched. Equal candidate counts do not make the selected EQ-on pipeline an unchanged-treatment factorial intervention.

After all configuration choices freeze, eleven named adapter policies per seed supply native readouts and two newly fitted linear heads. This gives 220 new heads and 660 named raw/adjusted readouts. Uniform-prior adjusted aliases repeat predictions. The CPU backend/thread count was chosen by fit-only throughput measurements.

A hash-bound analysis amendment was recorded before any confirmation scores. It retains four fixed-treatment factorial contrasts and adds two selected stronger-control contrasts to one six-member primary family; two selected EQ pipeline effects are secondary. Supplied timestamps and a preserved zero-confirmation counter establish recorded ordering, without external time attestation.

The original audit verifies all 320 adapter and 220 linear-head states. A separate local weight-free replay verifies all 320 saved tuning captures, 660 confirmation readouts and 330 decision-offset curves with zero model-state loads. It independently regenerates the clean development split, rank and thinning; it does not recompute feature extraction or fit-scaler estimation.

Tiny ImageNet | EQ and stronger loss controls | confirmation results

Method / readout	Macro-F1 (%)	Tail F1 (%)	Accuracy (%)
Selected CE + LA	56.68 +/- 0.10	46.40 +/- 0.36	55.93 +/- 0.26
Selected Balanced Softmax	55.77 +/- 0.23	47.56 +/- 0.47	56.44 +/- 0.21
Selected cached-feature BCL adaptation	50.87 +/- 0.31	35.62 +/- 0.57	53.24 +/- 0.26
Natural sampler, EQ off; balanced objective	53.75 +/- 0.19	46.37 +/- 0.40	54.98 +/- 0.18
Natural sampler, fixed EQ .8/.05	53.75 +/- 0.19	46.37 +/- 0.40	54.98 +/- 0.18
Class-uniform sampler, EQ off	53.75 +/- 0.21	46.31 +/- 0.40	55.00 +/- 0.19
Class-uniform sampler, fixed EQ .8/.05	53.75 +/- 0.21	46.31 +/- 0.40	55.00 +/- 0.19

Entries are mean +/- sample SD over ten optimization seeds. Tail groups use the fixed fit-frequency rank. Readout aliases and differing target priors are disclosed in the study notes.

Tiny ImageNet | EQ and stronger loss controls: among the listed confirmation readouts, Selected CE + LA has the largest mean macro-F1, 56.68% (sample SD 0.10 percentage points). This ranking is descriptive; the declared paired comparisons supply the inferential evidence.

Tiny ImageNet | EQ and stronger loss controls | paired comparisons

Paired contrast	Effect and 95% CI (pp)	Exact p	Study Holm p
Fixed EQ on minus off, natural sampler	+0.0003 [-0.0136, +0.0142]	0.962891	1.000000
Fixed EQ on minus off, class-uniform sampler	+0.0019 [-0.0077, +0.0114]	0.625000	1.000000
Class-uniform minus natural sampler, EQ off	-0.0000 [-0.1025, +0.1024]	1.000000	1.000000
Fixed EQ effect: class-uniform minus natural	+0.0016 [-0.0147, +0.0178]	0.830078	1.000000
Selected Balanced Softmax minus selected CE + LA	-0.91 [-1.09, -0.72]	0.001953	0.011719
Selected BCL adaptation minus selected Balanced Softmax	-4.90 [-5.12, -4.69]	0.001953	0.011719

Intervals use the paired-t rule and are not simultaneous confidence intervals. Exact p-values enumerate all 1,024 sign assignments. Holm adjustments apply to the declared primary family.

The declared family contains 6 paired comparisons. 2 reach exact sign-flip Holm p <= 0.05 within this study (0 positive and 2 negative contrasts in the stated subtraction direction). The table retains every comparison, including undetected effects.

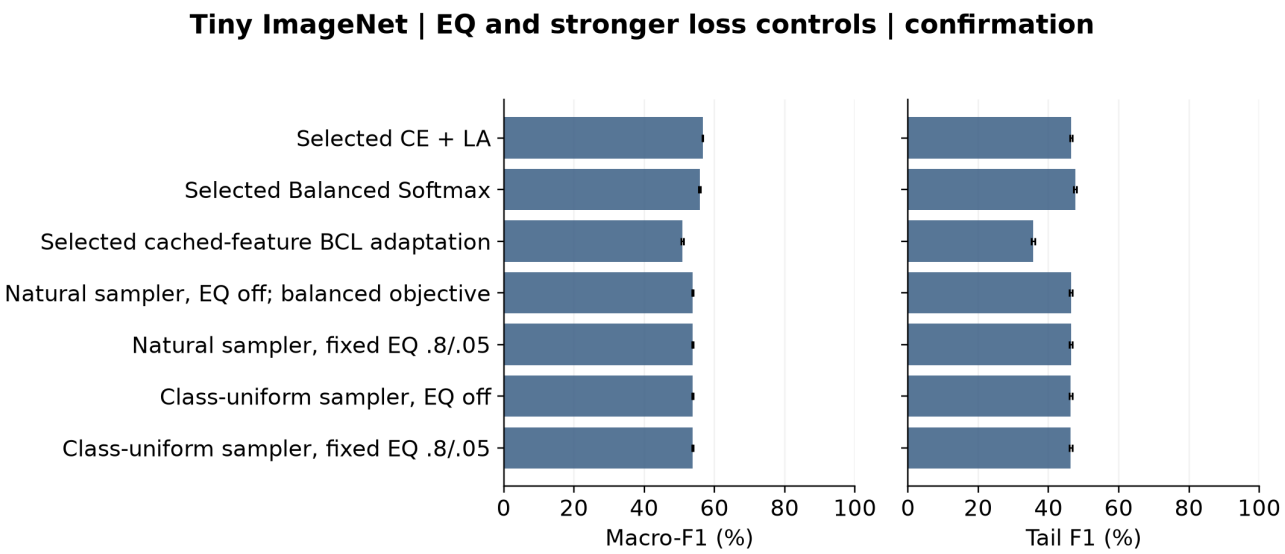
The stronger-control contrasts compare complete validation-selected pipelines; their learning rates can differ. All fixed factorial contrasts use the same loss target and the same prespecified EQ treatment. These statements do not establish a full GAN-versus-ViT causal ablation.

Adaptive EQ pipeline contrast	Effect (pp)	Paired 95% CI (pp)	Exact p
Selected natural EQ on minus off	+0.0003	[-0.0136, +0.0142]	0.962891
Selected class-uniform EQ on minus off	-0.0105	[-0.0386, +0.0176]	0.337891

Secondary adaptive-pipeline effects use separately selected EQ coefficients/onsets. Exact p-values are unadjusted; these two effects are outside the six-member primary family. They do not define the fixed-treatment factorial interaction.

Selected representation	Native head	Natural refit head	ROS refit head
Selected CE + LA	56.68 +/- 0.10	55.66 +/- 0.27	52.34 +/- 0.24
Selected Balanced Softmax	55.77 +/- 0.23	55.80 +/- 0.29	52.44 +/- 0.26
Selected BCL adaptation	50.87 +/- 0.31	54.29 +/- 0.24	52.81 +/- 0.21

Secondary confirmation macro-F1 (%), mean +/- sample SD. Natural and ROS readouts fit new linear classifiers for 2,000 updates on each fixed representation; ROS resamples actual fit vectors. Head changes can alter descriptive model ranking. These readouts are not additional independent encoder fits.



Mean +/- sample SD over ten optimizer seeds; tail groups fixed by fit rank. Named readouts can share fitted models.

Tiny ImageNet | encoder and partial-tuning controls | methodology

Six arms combine frozen DeiT-tiny, last-two-block DeiT-tiny tuning, or frozen DINOv2 ViT-S/14 with CE or Balanced Softmax. Frozen models train linear heads. Partial DeiT tunes the final two blocks, final norm and classifier; the first ten blocks stay frozen in evaluation mode.

All original 64x64 images use the same deterministic RGB, bicubic resize/crop to 224, and ImageNet normalization. Enlarging images adds no observed detail. This study introduces no stochastic image augmentation.

The FP16 frozen DeiT prefix has a train-only reconstruction-error gate. Its final trainable block computes only the CLS query/output while retaining all keys/values; zero-dropout outputs and gradients are independently checked against the complete block. DINO uses its own fit-only feature scaler.

Partition	Rows	Use
Full source	110000	100,000 development plus 10,000 official evaluation images
Imbalanced fit	15268	Same documented fixed image partition as EQ
Tuning	4000	Learning-rate and checkpoint selection
Calibration	4000	Decision adjustment within checkpoint/LR selection
Confirmation	20000	Primary calibrated macro-F1 after complete global freeze
Official evaluation	10000	Exploratory only

Dedicated tuning seed 901 evaluates two learning rates for each of six arms, followed by ten final seeds. Every fit completes 15 epochs, batch 128, 120 updates per epoch and 1,800 total updates. AdamW uses warmup/cosine decay, weight decay 0.01 and clipping at 1.

CE selects a calibration-set decision offset at every epoch and ranks checkpoints by adjusted tuning macro-F1. Consequently calibration is development data in this study. All six learning rates and all sixty final fit receipts are verified before any confirmation or official-evaluation outcome is opened.

Frozen head learning rates are 0.001/0.003; partial encoder rates are 0.00001/0.00003 with a 100-fold head multiplier. Completed training budgets are retained even when an earlier checkpoint wins. Balanced Softmax has a uniform effective prior, so its confirmation LA view repeats raw predictions.

Independent numeric replay covers sixty final models, 180 named readouts and 72 tuning/final development captures. A separate read-only audit loads all 72 actual checkpoint states and verifies tensor inventories, trainable/frozen eligibility, paired initialization, AdamW moments/groups and all 1,800 steps. The portable bundle loads zero states and does not reconstruct encoder inference.

Tiny ImageNet | encoder and partial-tuning controls | confirmation results

Method / readout	Macro-F1 (%)	Tail F1 (%)	Accuracy (%)
Frozen DeiT / CE + LA	62.24 +/- 0.11	57.03 +/- 0.36	62.28 +/- 0.13
Frozen DeiT / Balanced Softmax	61.74 +/- 0.08	57.24 +/- 0.17	62.12 +/- 0.06
Last-two-block DeiT / CE + LA	62.83 +/- 0.12	57.75 +/- 0.33	62.66 +/- 0.22
Last-two-block DeiT / Balanced Softmax	62.52 +/- 0.15	58.18 +/- 0.23	62.76 +/- 0.12
Frozen DINOv2 / CE + LA	74.59 +/- 0.16	69.24 +/- 0.33	74.50 +/- 0.17
Frozen DINOv2 / Balanced Softmax	73.41 +/- 0.13	68.53 +/- 0.23	73.83 +/- 0.13

Entries are mean +/- sample SD over ten optimization seeds. Tail groups use the fixed fit-frequency rank. Readout aliases and differing target priors are disclosed in the study notes.

Tiny ImageNet | encoder and partial-tuning controls: among the listed confirmation readouts, Frozen DINOv2 / CE + LA has the largest mean macro-F1, 74.59% (sample SD 0.16 percentage points). This ranking is descriptive; the declared paired comparisons supply the inferential evidence.

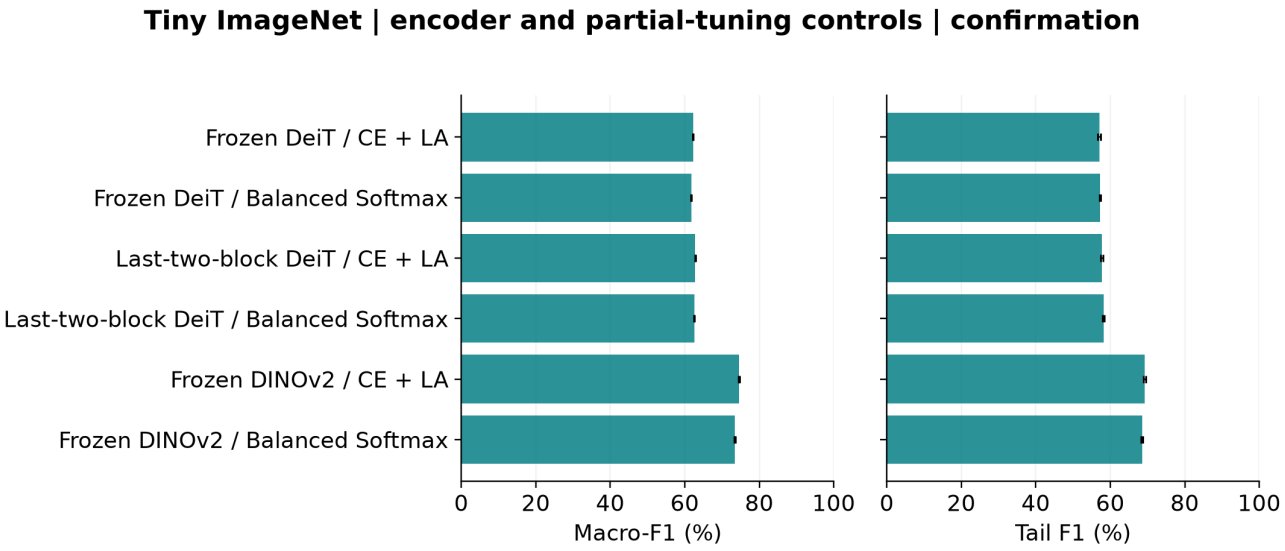
Tiny ImageNet | encoder and partial-tuning controls | paired comparisons

Paired contrast	Effect and 95% CI (pp)	Exact p	Study Holm p
Last-two-block DeiT CE + LA minus frozen DeiT CE + LA	+0.59 [+0.47, +0.72]	0.001953	0.009766
Frozen DINO CE + LA minus frozen DeiT CE + LA	+12.35 [+12.19, +12.52]	0.001953	0.009766
Balanced Softmax minus CE + LA, frozen DeiT	-0.50 [-0.59, -0.41]	0.001953	0.009766
Balanced Softmax minus CE + LA, last-two-block DeiT	-0.31 [-0.44, -0.19]	0.001953	0.009766
Balanced Softmax minus CE + LA, frozen DINO	-1.18 [-1.26, -1.11]	0.001953	0.009766

Intervals use the paired-t rule and are not simultaneous confidence intervals. Exact p-values enumerate all 1,024 sign assignments. Holm adjustments apply to the declared primary family.

The declared family contains 5 paired comparisons. 5 reach exact sign-flip Holm p <= 0.05 within this study (2 positive and 3 negative contrasts in the stated subtraction direction). The table retains every comparison, including undetected effects.

DINO versus DeiT compares model packages that differ in pretraining, width, patch size and capacity. Partial versus frozen DeiT changes trainable parameters and optimization. Neither contrast isolates architecture alone; the partial model is not full end-to-end fine-tuning. Reported wrapper parameter counts do not represent the whole pretrained backbone.



Mean +/- sample SD over ten optimizer seeds; tail groups fixed by fit rank. Named readouts can share fitted models.

DBpedia-14 | contextual and selective-tuning controls | methodology

Thirteen text arms retrain random mean-pooled controls and official pretrained BERT controls on the new partitions. Seven predictive frozen/mean arms use fifty epochs; matched frozen CLS, last-one-block and last-two-block BERT each pair CE and Balanced Softmax under the same three-epoch batch lists.

The fifty-epoch BERT representation is attention-masked contextual mean pooling with a fit-only scaler. The matched/partial group uses final CLS and its own shared fit-only scaler. The random mean baseline uses a fit-only 50,000-word vocabulary, a 128-word cap and a randomly initialized embedding-bag model. Word caps and WordPiece positions are different text units.

Natural sampling uses the empirical fit distribution; class-uniform sampling uses replacement without an extra class weight. Balanced Softmax uses naturally sampled CE(logits + log(fit counts)) and evaluates raw logits with uniform effective prior. Natural CE/EQ alone receive calibration-selected LA. The text EQ coefficient is 0.05 throughout all fifty adapter epochs; the image fixed-reference treatment starts at fraction 0.8.

The matched frozen CLS control uses CPU float32; selective tuning uses MPS float16 autocast with float32 master weights. Activating transformer dropout is part of the partial regime. The prospective CLS-only final-layer computation preserves all K/V tokens and is output/gradient equivalent at zero dropout; training dropout has the same marginal law with changed RNG ordering.

Partition	Rows	Use
Full native source	630000	560,000 development and 70,000 native test records
Canonical development	559860	Normalized full-record/content exact deduplication before partitioning
Long-tail fit	106232	14 classes; nominal 100:1 thinning; fit-only vocabulary/scalers
Tuning	28000	2,000 records per class; global LR selection only
Calibration	28000	2,000 per class; freeze CE decision offset before confirmation inference
Confirmation	56000	4,000 per class; primary balanced train-derived holdout
Native test	70000	Exploratory for frozen/mean arms only

Two full-budget learning-rate candidates per arm at dedicated seed 9 give 26 tuning fits. One global method learning rate is frozen before final seeds 10-19. Tune-only macro-F1 selects final-epoch candidates; all thirteen choices and all 130 final fits pass a whole-study gate before confirmation.

The fifty-epoch methods complete 5,200 updates; the matched/partial methods complete 318 updates on identical three-epoch fit coverage. Length buckets remove masked padding while keeping source IDs/valid tokens. Batch/backend scheduling was selected with fit-only throughput pilots.

Each CE calibration choice is persisted before its confirmation inference. All 190 named confirmation readouts and 100 exploratory native readouts retain exact labels, row IDs/ranks, logits, decision priors/tau and histories. Named raw/adjusted views are not additional independent encoder training fits.

Independent replay verifies 26 global tuning captures, 130 final cells, source-row binding, all metrics and the eight-member primary family. Its original audit loads all 156 candidate/final checkpoint states; the portable audit loads zero and explicitly declares 156 omitted states.

DBpedia-14 | contextual and selective-tuning controls | confirmation results

Method / readout	Macro-F1 (%)	Tail F1 (%)	Accuracy (%)
Frozen BERT mean / CE + LA	98.69 +/- 0.01	98.45 +/- 0.04	98.69 +/- 0.01
Frozen BERT mean / Balanced Softmax	98.70 +/- 0.01	98.48 +/- 0.02	98.70 +/- 0.01
Frozen BERT mean / natural EQ + LA	98.70 +/- 0.02	98.47 +/- 0.02	98.70 +/- 0.02
Frozen BERT mean / class-uniform CE	98.54 +/- 0.02	98.27 +/- 0.04	98.54 +/- 0.02
Frozen BERT mean / class-uniform EQ	98.56 +/- 0.01	98.29 +/- 0.02	98.56 +/- 0.01
Random mean-pooled text / CE + LA	82.49 +/- 2.26	69.98 +/- 3.58	82.65 +/- 2.18
Random mean-pooled text / Balanced Softmax	84.19 +/- 1.99	71.05 +/- 3.28	84.27 +/- 1.89
Matched frozen BERT CLS / CE + LA	97.18 +/- 0.08	96.16 +/- 0.21	97.17 +/- 0.08
Matched frozen BERT CLS / Balanced Softmax	97.21 +/- 0.08	96.22 +/- 0.21	97.21 +/- 0.08
Last-one-block BERT CLS / CE + LA	98.97 +/- 0.03	98.77 +/- 0.05	98.97 +/- 0.03
Last-one-block BERT CLS / Balanced Softmax	98.96 +/- 0.03	98.79 +/- 0.05	98.96 +/- 0.03
Last-two-block BERT CLS / CE + LA	99.05 +/- 0.02	98.87 +/- 0.05	99.05 +/- 0.02
Last-two-block BERT CLS / Balanced Softmax	99.04 +/- 0.01	98.88 +/- 0.03	99.03 +/- 0.01

Entries are mean +/- sample SD over ten optimization seeds. Tail groups use the fixed fit-frequency rank. Readout aliases and differing target priors are disclosed in the study notes.

DBpedia-14 | contextual and selective-tuning controls: among the listed confirmation readouts, Last-two-block BERT CLS / CE + LA has the largest mean macro-F1, 99.05% (sample SD 0.02 percentage points). This ranking is descriptive; the declared paired comparisons supply the inferential evidence.

DBpedia-14 | contextual and selective-tuning controls | paired comparisons

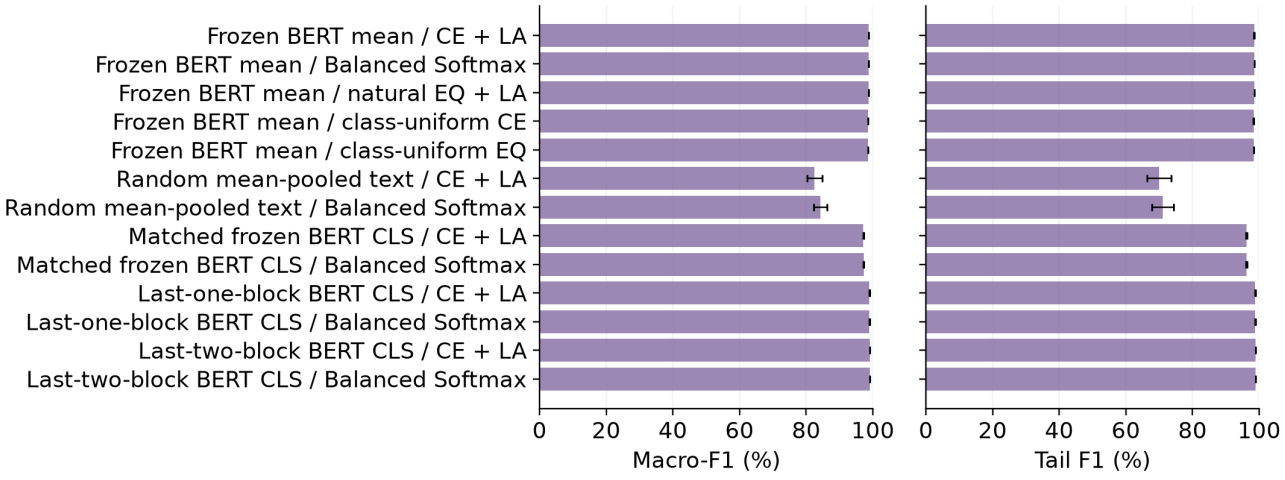
Paired contrast	Effect and 95% CI (pp)	Exact p	Study Holm p
Frozen BERT mean CE + LA minus random mean CE + LA	+16.19 [+14.57, +17.81]	0.001953	0.015625
Frozen BERT mean Balanced Softmax minus CE + LA	+0.01 [+0.00, +0.03]	0.011719	0.035156
EQ minus CE, class-uniform frozen BERT mean	+0.02 [+0.01, +0.03]	0.011719	0.035156
Natural EQ + LA minus CE + LA, frozen BERT mean	+0.01 [+0.00, +0.03]	0.048828	0.048828
Last-one-block CE + LA minus matched frozen CLS CE + LA	+1.79 [+1.73, +1.85]	0.001953	0.015625
Last-two-block CE + LA minus matched frozen CLS CE + LA	+1.87 [+1.81, +1.94]	0.001953	0.015625
Last-one-block Balanced Softmax minus matched frozen CLS	+1.75 [+1.69, +1.81]	0.001953	0.015625
Last-two-block Balanced Softmax minus matched frozen CLS	+1.82 [+1.76, +1.88]	0.001953	0.015625

Intervals use the paired-t rule and are not simultaneous confidence intervals. Exact p-values enumerate all 1,024 sign assignments. Holm adjustments apply to the declared primary family.

The declared family contains 8 paired comparisons. 8 reach exact sign-flip Holm p <= 0.05 within this study (8 positive and 0 negative contrasts in the stated subtraction direction). The table retains every comparison, including undetected effects.

Each text arm selects its own learning rate. EQ-on/off and Balanced Softmax/CE contrasts therefore compare selected pipelines and can include learning-rate differences; they do not isolate a loss toggle. Selective tuning is compared to the matched three-epoch frozen CLS group. Its contrast includes dropout, precision and backend effects. The fifty-epoch mean-pooled BERT baseline is a predictive reference; different pooling and budgets preclude attributing that difference solely to encoder tuning. Text sampler objectives differ, so their EQ contrasts do not establish an unchanged-objective factorial interaction.

DBpedia-14 | contextual and selective-tuning controls | confirmation



Mean +/- sample SD over ten optimizer seeds; tail groups fixed by fit rank. Named readouts can share fitted models.

Covertype | tabular model and objective controls | methodology

Four families compare a natural CE MLP, effective-number-weighted MLP, a genuine FT-Transformer implementation and XGBoost histogram trees. This study retains the natural Covertype class frequencies and adds no synthetic observations.

Fit-only statistics standardize ten continuous variables. MLP/trees use all 54 original predictors; FT uses ten numeric feature tokens plus two categorical tokens that losslessly encode the four wilderness and forty soil indicators. Exact one-hot reconstruction is audited.

The FT implementation has numerical affine feature tokens, categorical embeddings/biases, CLS, and two attention/ReGLU blocks. It is an architecture-family implementation with disclosed settings, not the unmodified official RTDL code. Effective-number weighting applies throughout MLP fitting and is normalized to expected fit sample weight one.

Partition	Rows	Use
Full source	581012	All canonical UCI records; original row IDs retained
Fit	406704	Class-stratified 70% floor; transform and gradients
Tuning	58099	10% floor; raw-logit checkpoint/configuration selection
Calibration	58099	10% floor; decision-offset selection after global configuration freeze
Confirmation	58110	Per-class remainder; primary natural-prior evaluation

Dedicated tuning seed 9 evaluates two configurations per family, giving eight full-budget trials; ten final seeds give forty models and eighty raw/adjusted readouts. Neural models complete 50 AdamW epochs with five-epoch warmup/cosine decay; MLP batch 2048 gives 9,950 updates and FT batch 1024 gives 19,900 updates.

XGBoost completes 600 multiclass boosting rounds (4,200 trees); the tuning set selects the reported iteration among 50-round intervals through 600. Equal two-trial searches use model-specific budgets and do not claim equal FLOPs or wall time.

Raw-logit tuning macro-F1 selects configurations/checkpoints. All eight tuning captures and four selected family configurations freeze before calibration/confirmation. Macro-F1 decision offsets use the effective weighted prior for the weighted MLP and the fit-frequency prior for the other families.

The fresh confirmation partition overlaps 8,669 rows of the previous study's test partition. Shared CPU/MPS load and checkpoint resumption affect invocation timings; retained training time and actual phase wall time are separate records. Runtime records do not establish isolated efficiency.

The original metadata/checkpoint audit includes the scientific runner’s numeric self-replay. A separate verifier imports no scientific runner and recomputes all forty final prediction cells, eighty raw/adjusted readouts, forty eight-candidate decision-offset curves, scalar and per-class metrics with NumPy confusion matrices crosschecked against scikit-learn. The portable final prediction replay loads zero model weights and does not reconstruct inference.

Covertype | tabular model and objective controls | confirmation results

Method / readout	Macro-F1 (%)	Tail F1 (%)	Accuracy (%)
Natural CE MLP + LA	91.54 +/- 0.15	85.46 +/- 0.55	94.46 +/- 0.10
Effective-number MLP + LA	88.04 +/- 0.17	79.03 +/- 0.45	91.74 +/- 0.10
FT-Transformer + LA	89.81 +/- 0.33	82.87 +/- 0.62	93.65 +/- 0.18
XGBoost + LA	93.10 +/- 0.05	88.73 +/- 0.16	94.47 +/- 0.05

Entries are mean +/- sample SD over ten optimization seeds. Tail groups use the fixed fit-frequency rank. Readout aliases and differing target priors are disclosed in the study notes.

Covertype | tabular model and objective controls: among the listed confirmation readouts, XGBoost + LA has the largest mean macro-F1, 93.10% (sample SD 0.05 percentage points). This ranking is descriptive; the declared paired comparisons supply the inferential evidence.

Covertype | tabular model and objective controls | paired comparisons

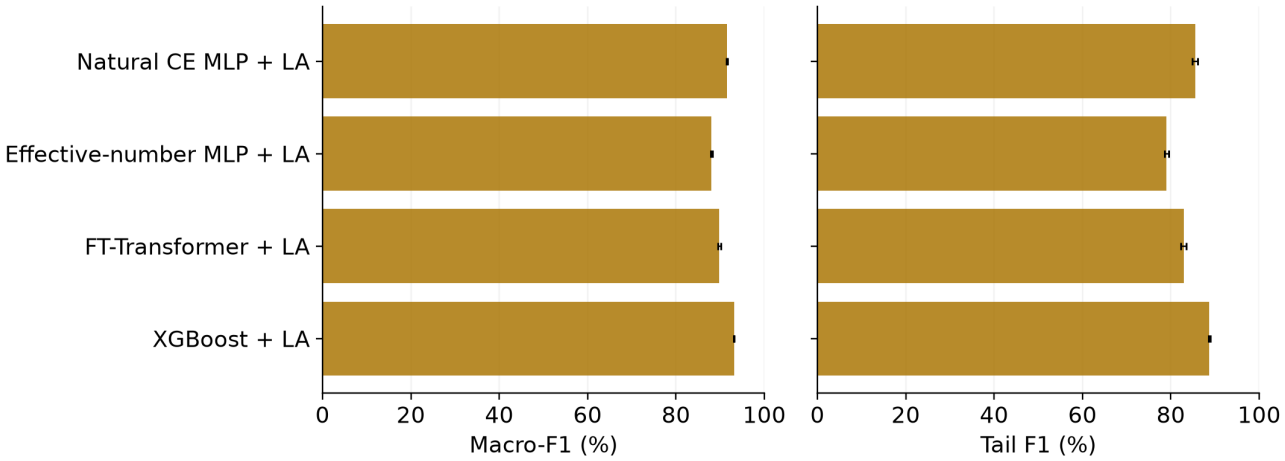
Paired contrast	Effect and 95% CI (pp)	Exact p	Study Holm p
FT-Transformer + LA minus natural CE MLP + LA	-1.73 [-2.00, -1.46]	0.001953	0.005859
XGBoost + LA minus natural CE MLP + LA	+1.57 [+1.46, +1.68]	0.001953	0.005859
Effective-number MLP + LA minus natural CE MLP + LA	-3.50 [-3.63, -3.37]	0.001953	0.005859

Intervals use the paired-t rule and are not simultaneous confidence intervals. Exact p-values enumerate all 1,024 sign assignments. Holm adjustments apply to the declared primary family.

The declared family contains 3 paired comparisons. 3 reach exact sign-flip Holm p <= 0.05 within this study (1 positive and 2 negative contrasts in the stated subtraction direction). The table retains every comparison, including undetected effects.

The three primary effects compare adjusted FT, XGBoost and effective-number MLP to adjusted natural CE MLP. Natural-prior confirmation macro-F1 is primary. Uniform-prior reweighting is a secondary diagnostic on the same records, not another independent dataset.

Covertype | tabular model and objective controls | confirmation



Mean +/- sample SD over ten optimizer seeds; tail groups fixed by fit rank. Named readouts can share fitted models.

Global multiplicity sensitivity | 1 of 2

Study / contrast	Effect and 95% CI (pp)	Study Holm p	Global Holm p
equalization / Fixed EQ on minus off, natural sampler	+0.0003 [-0.0136, +0.0142]	1.000000	1.000000
equalization / Fixed EQ on minus off, class-uniform sampler	+0.0019 [-0.0077, +0.0114]	1.000000	1.000000
equalization / Class-uniform minus natural sampler, EQ off	-0.0000 [-0.1025, +0.1024]	1.000000	1.000000
equalization / Fixed EQ effect: class-uniform minus natural	+0.0016 [-0.0147, +0.0178]	1.000000	1.000000
equalization / Selected Balanced Softmax minus selected CE + LA	-0.91 [-1.09, -0.72]	0.011719	0.042969
equalization / Selected BCL adaptation minus selected Balanced Softmax	-4.90 [-5.12, -4.69]	0.011719	0.042969
vision / Last-two-block DeiT CE + LA minus frozen DeiT CE + LA	+0.59 [+0.47, +0.72]	0.009766	0.042969
vision / Frozen DINO CE + LA minus frozen DeiT CE + LA	+12.35 [+12.19, +12.52]	0.009766	0.042969
vision / Balanced Softmax minus CE + LA, frozen DeiT	-0.50 [-0.59, -0.41]	0.009766	0.042969
vision / Balanced Softmax minus CE + LA, last-two-block DeiT	-0.31 [-0.44, -0.19]	0.009766	0.042969
vision / Balanced Softmax minus CE + LA, frozen DINO	-1.18 [-1.26, -1.11]	0.009766	0.042969

Both adjusted p columns use the exact sign-flip test. Global correction is sensitivity analysis, not replacement of the prespecified study families. The paired-t p-values, all individual differences and raw p-values are retained in machine-readable evidence.

Global multiplicity sensitivity | 2 of 2

Study / contrast	Effect and 95% CI (pp)	Study Holm p	Global Holm p
text / Frozen BERT mean CE + LA minus random mean CE + LA	+16.19 [+14.57, +17.81]	0.015625	0.042969
text / Frozen BERT mean Balanced Softmax minus CE + LA	+0.01 [+0.00, +0.03]	0.035156	0.082031
text / EQ minus CE, class-uniform frozen BERT mean	+0.02 [+0.01, +0.03]	0.035156	0.082031
text / Natural EQ + LA minus CE + LA, frozen BERT mean	+0.01 [+0.00, +0.03]	0.048828	0.244141
text / Last-one-block CE + LA minus matched frozen CLS CE + LA	+1.79 [+1.73, +1.85]	0.015625	0.042969
text / Last-two-block CE + LA minus matched frozen CLS CE + LA	+1.87 [+1.81, +1.94]	0.015625	0.042969
text / Last-one-block Balanced Softmax minus matched frozen CLS	+1.75 [+1.69, +1.81]	0.015625	0.042969
text / Last-two-block Balanced Softmax minus matched frozen CLS	+1.82 [+1.76, +1.88]	0.015625	0.042969
tabular / FT-Transformer + LA minus natural CE MLP + LA	-1.73 [-2.00, -1.46]	0.005859	0.042969
tabular / XGBoost + LA minus natural CE MLP + LA	+1.57 [+1.46, +1.68]	0.005859	0.042969
tabular / Effective-number MLP + LA minus natural CE MLP + LA	-3.50 [-3.63, -3.37]	0.005859	0.042969

Both adjusted p columns use the exact sign-flip test. Global correction is sensitivity analysis, not replacement of the prespecified study families. The paired-t p-values, all individual differences and raw p-values are retained in machine-readable evidence.

Practical interpretation and limitations

Use the within-study paired comparisons and their uncertainty when choosing a pipeline for these fixed partitions. Descriptive score leaders are not evidence of universal superiority. Small, undetected or negative improvements are retained rather than removed from the report.

Strong calibrated CE and Balanced Softmax controls remain necessary when evaluating EQ or contrastive methods. Compare selective tuning against the matched frozen control. Evaluate tabular attention alongside boosted trees and conventional MLP objectives under the natural data distribution.

Image EQ and sampling: retain the fixed-treatment comparisons separately from the selected stronger-loss pipelines. Fixed EQ on minus off with natural sampling: +0.0003 pp, 95% paired interval [-0.0136, +0.0142], study Holm p=1.000000; no nonzero effect is detected; this does not establish equivalence. Fixed EQ on minus off with class-uniform sampling: +0.0019 pp, 95% paired interval [-0.0077, +0.0114], study Holm p=1.000000; no nonzero effect is detected; this does not establish equivalence. Class-uniform minus natural sampling with EQ off: -0.0000 pp, 95% paired interval [-0.1025, +0.1024], study Holm p=1.000000; no nonzero effect is detected; this does not establish equivalence. Fixed EQ-by-sampling interaction: +0.0016 pp, 95% paired interval [-0.0147, +0.0178], study Holm p=1.000000; no nonzero effect is detected; this does not establish equivalence. Selected Balanced Softmax minus selected CE + LA: -0.9065 pp, 95% paired interval [-1.0884, -0.7246], study Holm p=0.011719; a negative effect is detected. Selected BCL adaptation minus selected Balanced Softmax: -4.9042 pp, 95% paired interval [-5.1211, -4.6874], study Holm p=0.011719; a negative effect is detected. Selected native adjusted macro-F1 means are CE 56.680%, Balanced Softmax 55.774% and BCL adaptation 50.870%. Keep CE and Balanced Softmax as controls. The fixed natural/class-uniform policies target the same class-balanced objective; selected loss pipelines can also differ in their selected settings. BCL uses cached features and adapter-dropout views, rather than full-backbone image augmentation. New natural/ROS heads are diagnostic readouts, not independent encoder fits.

Image encoders: compare each loss with its matched encoder control. Last-two-block DeiT CE + LA minus frozen DeiT CE + LA: +0.5948 pp, 95% paired interval [+0.4717, +0.7179], study Holm p=0.009766; a positive effect is detected. Frozen DINOv2 CE + LA minus frozen DeiT CE + LA: +12.3526 pp, 95% paired interval [+12.1885, +12.5167], study Holm p=0.009766; a positive effect is detected. Balanced Softmax minus CE + LA for frozen DeiT: -0.4992 pp, 95% paired interval [-0.5856, -0.4128], study Holm p=0.009766; a negative effect is detected. Balanced Softmax minus CE + LA for last-two-block DeiT: -0.3133 pp, 95% paired interval [-0.4377, -0.1889], study Holm p=0.009766; a negative effect is detected. Balanced Softmax minus CE + LA for frozen DINOv2: -1.1835 pp, 95% paired interval [-1.2597, -1.1073], study Holm p=0.009766; a negative effect is detected. Adjusted CE macro-F1 means are frozen DeiT 62.237%, last-two-block DeiT 62.832% and frozen DINOv2 74.590%. DINOv2 versus DeiT changes the whole encoder package, including capacity and pretraining, so its contrast does not isolate an architecture or loss intervention. Partial versus frozen DeiT compares the declared selected pipelines. Experimental partition disjointness does not establish independence from ImageNet pretraining.

Text selective tuning: use the four declared partial-versus-matched-frozen comparisons to assess tuning. Last-one-block CE + LA minus matched frozen CLS CE + LA: +1.7910 pp, 95% paired interval [+1.7333, +1.8488], study Holm p=0.015625; a positive effect is detected. Last-two-block CE + LA minus matched frozen CLS CE + LA: +1.8717 pp, 95% paired interval [+1.8059, +1.9374], study Holm p=0.015625; a positive effect is detected. Last-one-block Balanced Softmax minus matched frozen CLS Balanced Softmax: +1.7495 pp, 95% paired interval [+1.6871, +1.8119], study Holm p=0.015625; a positive effect is detected. Last-two-block Balanced Softmax minus matched frozen CLS Balanced Softmax: +1.8224 pp, 95% paired interval [+1.7617, +1.8830], study Holm p=0.015625; a positive effect is detected. Matched frozen CLS macro-F1 means are CE + LA 97.178% and Balanced Softmax 97.213%. Each arm selects its own learning rate; the prescribed tuning groups also change backend, precision and dropout. These effects therefore describe selected pipelines, not an isolated trainability toggle. The fifty-epoch mean-pooled BERT references use different pooling and budgets from the matched three-epoch CLS comparison. Do not substitute those references for the matched frozen control or interpret a non-detection as equivalence.

Tabular models: retain boosted trees and conventional CE MLP as predictive controls under the natural data distribution. Adjusted XGBoost minus adjusted natural CE MLP: +1.5679 pp, 95% paired interval [+1.4580, +1.6778], study Holm p=0.005859; a positive effect is detected. Adjusted local FT-Transformer minus adjusted natural CE MLP: -1.7312 pp, 95% paired interval [-1.9991, -1.4634], study Holm p=0.005859; a negative effect is detected. Adjusted effective-number MLP minus adjusted natural CE MLP: -3.4998 pp, 95% paired interval [-3.6294, -3.3701], study Holm p=0.005859; a negative effect is detected. Adjusted macro-F1 means are XGBoost 93.104% and CE MLP 91.536%. Effective-number versus CE MLP means are macro recall 94.824% versus 91.864%, uniform-prior diagnostic macro-F1 94.816% versus 91.953%, native-prior macro-F1 88.037% versus 91.536% and tail F1 79.035% versus 85.459%. XGBoost macro-F1 is 93.257% raw and 93.104% adjusted; its macro recall is 92.462% raw and 94.003% adjusted, and its uniform-prior diagnostic macro-F1 is 92.552% raw and 94.050% adjusted. Preserve the declared adjusted primary endpoint and both raw/adjusted views; calibration-set decision selection need not improve confirmation macro-F1. Uniform-prior diagnostics reweight the same records, not a separate dataset. The local FT result does not rank all transformer configurations. Evaluate future objectives or decision policies on new prespecified data rather than selecting them from confirmation results.

The official DeiT checkpoint is ImageNet-pretrained, while Tiny ImageNet is sampled and downsampled from ImageNet. Item-level pretraining exclusion has not been established. Disjoint experimental partitions therefore do not establish independence from pretraining; this is a source-derived limitation, not a measured exact-item overlap rate.

The ten-seed intervals condition on the fixed source partitions and selected configurations. They do not cover hyperparameter-selection uncertainty or external dataset/cohort uncertainty.

The image studies reuse Tiny ImageNet source rows; historical models may have seen the new confirmation images. Covertype is a previously studied, spatially dependent source. New partitions protect this round from its own selection, but do not provide an independent dataset, geographic cohort or external replication.

BERT was pretrained on Wikipedia and DBpedia abstracts derive from Wikipedia. Normalized exact deduplication does not remove pretraining, entity or paraphrase overlap. Encoder-package contrasts also change pretraining/capacity, while partial tuning changes dropout, optimizer and sometimes precision/backend. Results do not isolate transformer architecture alone.

Ten optimization seeds quantify initialization/training variation conditional on fixed partitions. Their intervals exclude data resampling, dependence across entities/locations and foundation-model pretraining uncertainty.
Non-significance does not establish equivalence; a degenerate zero-variance paired difference uses the declared finite statistics convention.

The studies do not conduct a fully matched GAN-versus-ViT intervention, full BCL image-augmentation reproduction, full backbone fine-tuning or probability-calibration reliability study. Optional SMOTE graphs and extra synthetic observations are absent from these primary follow-ups. Wrapper parameter counts must not be described as complete pretrained backbone capacity.

Reported training/phase timings reflect a shared machine, selected backends and some checkpoint resumption. They document resource use, not isolated throughput superiority or equal compute across model families.

Reproducibility and artifact identities

The public source and study documentation specify acquisition, canonicalization, source/model revisions, fit-only transformations, training grids, budgets and inference gates. Portable evidence retains scalar/class results, labels, source IDs/ranks, predictions, selected configuration records, histories and original audit receipts.

Full raw numeric portable archives remain local in the supplied evidence archive. Git contains smaller derived results, summaries, audit receipts, catalogs, report and source; the multi-gigabyte raw prediction blobs are not uploaded to Git. Image prediction row order is bound to saved partition indices and source labels rather than explicit per-prediction source-ID arrays.

To reproduce the current EQ analysis, run scripts/apply_followup_amendment.py on equalization/public_amendment_inputs with the frozen scientific source and counter snapshot. The historical runner aggregate remains archived; the amendment helper recreates the six-member primary and two secondary analyses.

Run each package numeric verifier in its documented weight-free mode; it must declare zero loaded model states. Original actual-state verification requires the omitted working checkpoints, dense inputs and pinned weights. Then run scripts/analyze_followup_comparisons.py on the four complete aggregates and this assembler before the report builder.

Scientific source hashes, protocol fingerprints, aggregate hashes, separate original/portable audit receipts and the global analysis identity are retained below. Credentials, private machine paths and operational logs are excluded from publication. Supplied record timestamps do not constitute external timestamp attestation.

Artifact role	SHA256
equalization scientific source	874e9b744e27f99813d729a16b4dbb4cda5571bf0219f3ee1ea0351311ac2705
equalization protocol fingerprint	0dfd887a0676a0a2d7803a6fbd7b6db073421136968260def67c3335752c00c1
equalization aggregate	d01fa2c535892c6d929afb4461b0abfd020c7624eeb0226f41ec57f64c434d27
equalization independent_verification	4a16321367f3f798e9cda4fe7afe7765eb80151bec2e50950eb33c70c0669cb2
vision scientific source	20711dd400a1e2c1c926344c91aec8c9bc1157546078a1cf6bf0c67665c6be3f
vision protocol fingerprint	f548c396fcf87f4e70aaa6d39da1b0f91f918de3934dc54ccec91e9927af4912
vision aggregate	bfb260a879ffdb1dfc75dc9eb7a269a5bc70765c7fc322ae6a4584a7f88b3539
vision independent_verification	1bc83746f84a34f690ab0eaabb0dfe7d363ae35acada62f1dad3dc49a6be3c03
vision checkpoint_verification	831c11c8a1746ea44dd7ef52d005a99046cc91976d9a34a4494b8fc668e4f0c3
text scientific source	8f05dc4a87813a9ffa4c1ac1b1de322cb0315cdb0c3237914bcefaa0cb12b83c
text protocol fingerprint	cd3fcb35f1b336cf7fab7c220c6f8edd1313a123d4e49eb145252d7a5bdd5342
text aggregate	51d33f599706786a918bf1a2c32f748f47185d7fb5ada74eb90cf44365f5a060
text independent_verification	d4b967cfa6815c03dd9de2114adb151277fa7658c6e1c7c3fd674e1f9f6087e9
tabular scientific source	a75df97f8351324e047bf19b298d9e421e489a6f9ffc679248580199c7869a67
tabular protocol fingerprint	f8d9bd1ce23e7dced07a4223b9888b48dd75f3f74ae902d83117535a079d7932
tabular aggregate	a08f2f433bfe51740949ccc0e1057c431a6c65ba50c2d7fbc02afb2e480914e7
tabular independent_verification	2e01d2e80f6197e5776acf9ff6c3ef73aafebf2bd1f2bfaab6e7bcf15287767a
tabular checkpoint_verification	97800349a47fa55fff35e080f8252ea65b555b1315648b855a5935bb505c948e

Artifact role	SHA256
Global 22-comparison analysis file	d7de2dfa20fc0baf7a60a83309f5a5a833b7ccc75539b6d23a37dcaa3d124ea6
Global comparison helper source	faec1269a4a916c8ace36f9333ce149ef19131db6f557188f6744b0055f47953

Dense inputs, pretrained feature caches, model/optimizer states, credentials and private operational logs are excluded from public evidence. Stored prediction records and split labels permit independent numeric replay; omitted training assets retain hash identities. The report does not claim to replay all encoder inference from the portable archive.

Primary method and dataset references

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